ELECTROSTATICS

1. Draw the field lines when the charge density of the sphere is (i) positive (ii) negative.

   [Delhi 2008]

2. Draw the equipotential surfaces due to an electric dipole. Locate the points where the potential due to the dipole is zero.

   [All India 2009C, 2011C, 2013]

3. Draw a graph of electric fields $E(r)$ with distance r from the centre of the shell for $0 \leq r \leq \infty$.

   [Delhi 2009]

4. Two charges of 5μC and -5μC are placed at points A and B 2 cm apart. Depict an equipotential surface of the system.

   [Delhi 2013C]

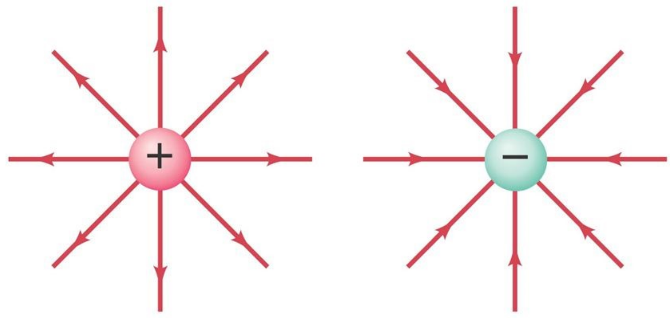
5. Draw equipotential surfaces:
   (i) in case of a single point charge and
   (ii) in a constant electric field in Z-direction. [All India 2016]

6. Draw a graph of E versus r for r >> a. [All India 2017]

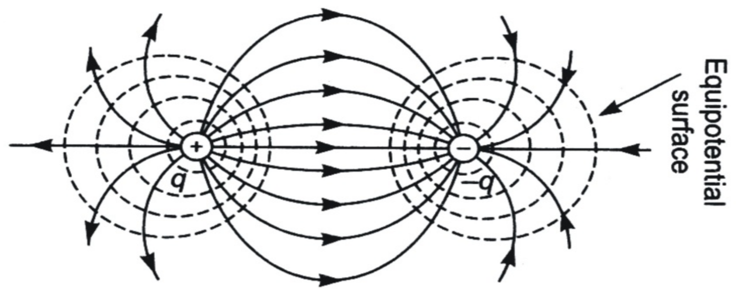
SOLUTIONS

1. Electric field lines when the charged density of the sphere is,

   (i) Positive (ii) Negative



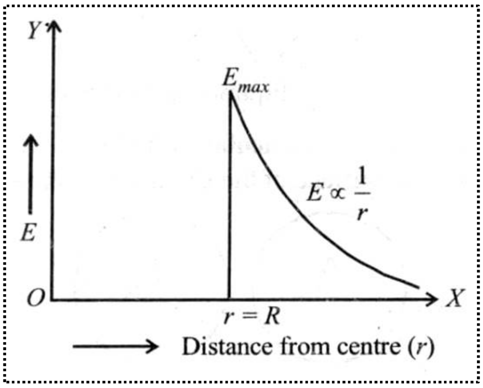
2. Equipotential surfaces due to an electric dipole.



Equipotential surfaces due to electric dipole

Potential due to the dipole is zero at the line bisecting the dipole length.

3. Graph of variation of electric field intensity E(r) with distance r from the centre for shell $0 \leq r < \infty$.



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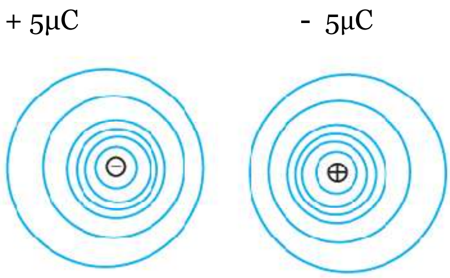
4. The equipotential surface of the system is as shown:



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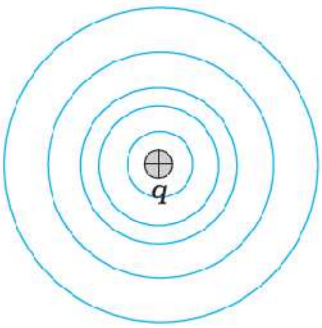
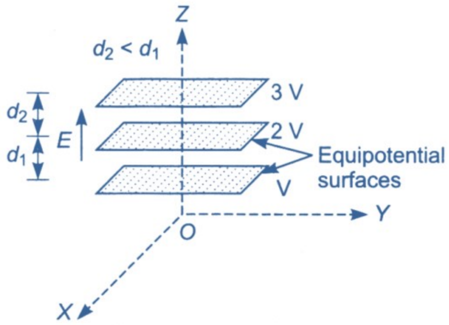
5. (i) Equipotential surface in case of single point charge.



(ii) Equipotential surface in a constant electric field in z-direction.



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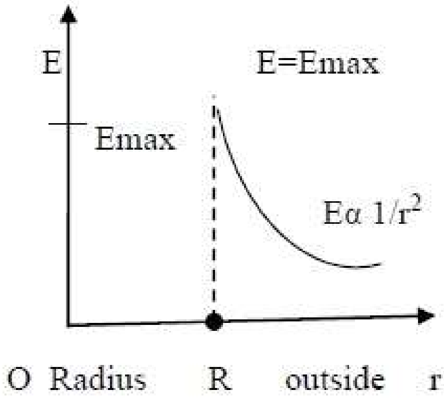
6. (b) Graph of E versus r for r >> a

$\because$ $E = \frac{p}{4\pi\varepsilon_0}\frac{2r}{r^4} = \frac{2p}{4\pi\varepsilon_0 r^3}$

$\therefore$ $E \propto \frac{1}{r^3}$



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7. Plot a graph showing the variation of resistivity with temperature for a metallic conductor.

[Delhi 2008]

8. Draw a graph showing the variation of resistivity with temperature for nichrome.

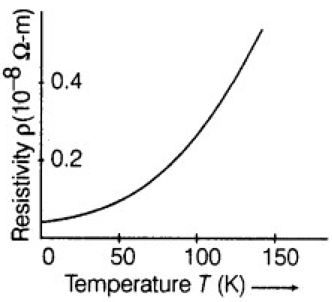
[All India 2013C]

9. A cell of emf ‘E’ and internal resistance ‘r’ is connected across a variable load resistor R. Draw the plots of the terminal voltage V versus (i) R and (ii) the current I.

[Delhi 2015]

SOLUTIONS

7. Resistivity of a conductor is defined as the resistance offered by unit length and unit area of cross-section of material of the conductor.



The temperature dependence of resistivity of a metal can be obtained from,
$r = r_0[1 + \alpha(T - T_0)]$
where r and r_0 are the resistivity at temperature T and T_0 respectively and α is called temperature coefficient of resistivity.
The value of α is positive, shows that resistivity increases with increases in temperature.

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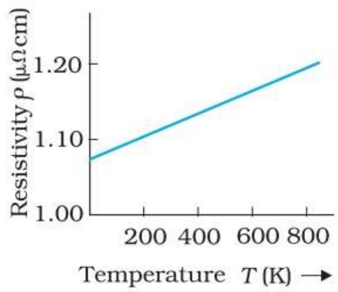
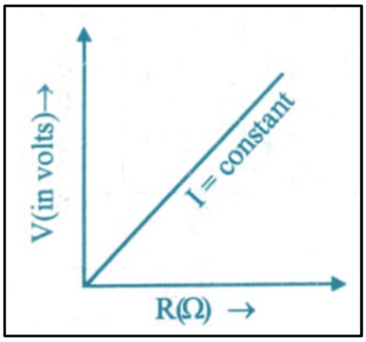
8. Graph of variation of resistivity with temperature for nichrome.

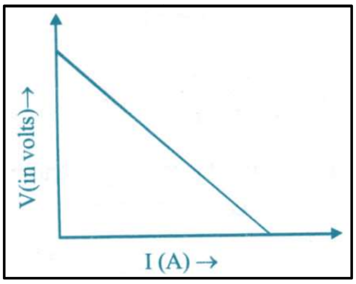
Property of nichrome used to make standard resistance coils: It has low temperature coefficient of resistance.



9. (i) Graph between terminal voltage V and resistance (R)
In the situation when no current is drawn from the cell then
$V = E$ ($\because V = E - Ir$ and $I = 0$)



(ii) Graph between terminal voltage (V) and current (I)



MAGNETISM

10. Draw a schematic sketch of a cyclotron.
[Delhi 2008, 2011C, 2012, All India 2013,2014]

11. Draw a labelled diagram of a moving coil galvanometer.
[Foreign 2012, All India 2014]

12. Sketch the magnetic field lines for a finite solenoid.
[Foreign 2010]

13. Draw magnetic field lines when a (i) diamagnetic, (ii) paramagnetic substance is placed in an external magnetic field.
[Delhi 2010]

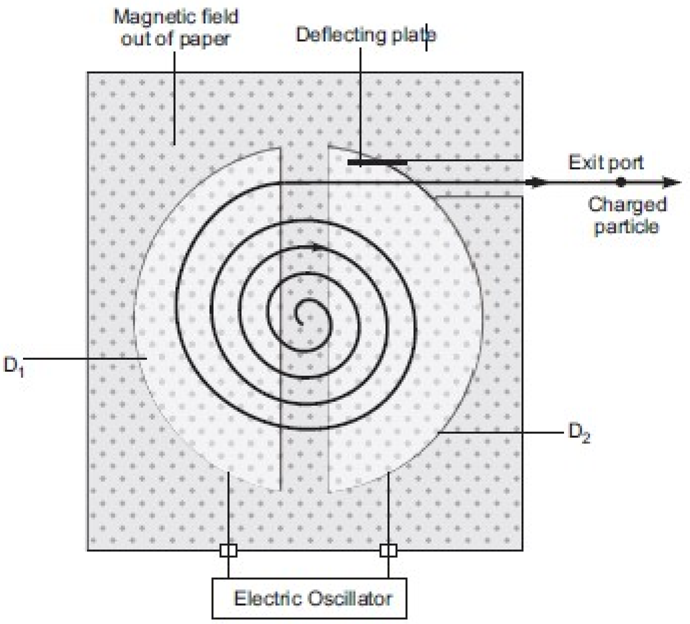
14. Draw the magnetic field lines due to a current carrying loop.
[Foreign 2010, Delhi 2013C]

15. The current flowing through an inductor of self inductance L is continuously increasing. Plot a graph showing the variation of
(i) Magnetic flux versus the current
(ii) Induced emf versus dI/dt
(iii) Magnetic potential energy stored versus the current. [Delhi 2014]

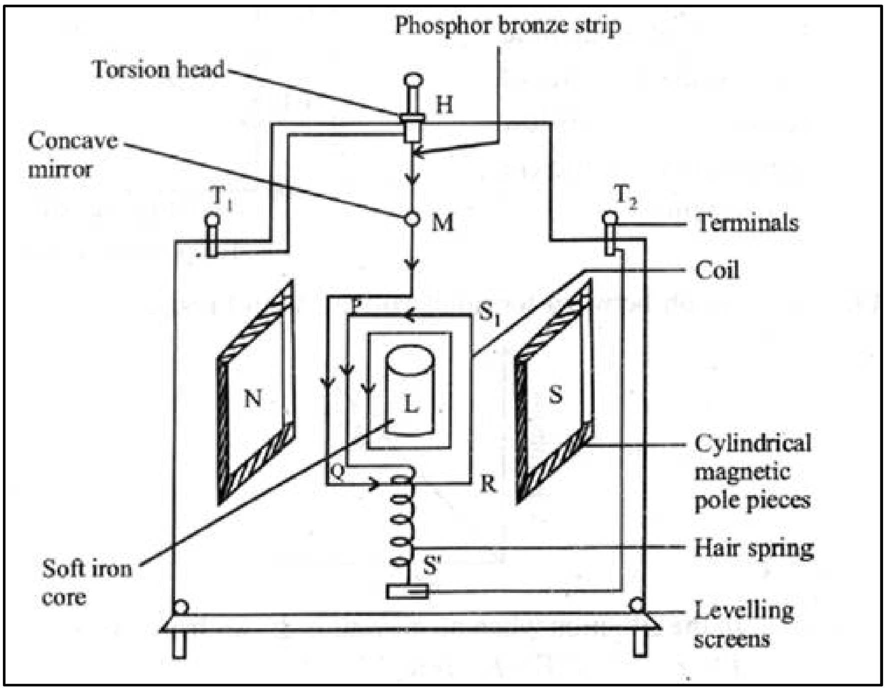
16. Draw the magnetic field lines due to a circular wire carrying current *I*.
[All India 2016]

SOLUTIONS

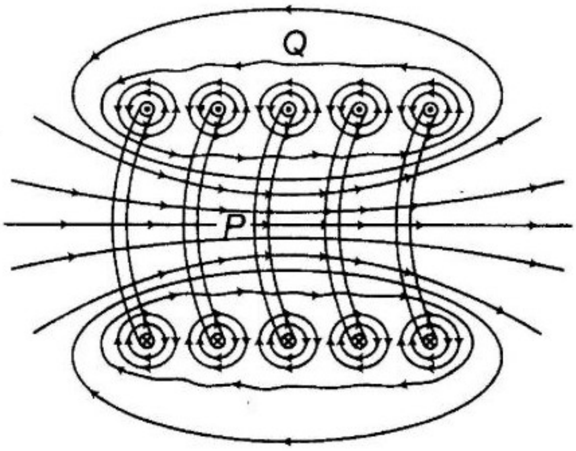
10. Schematic sketch of a Cyclotron.



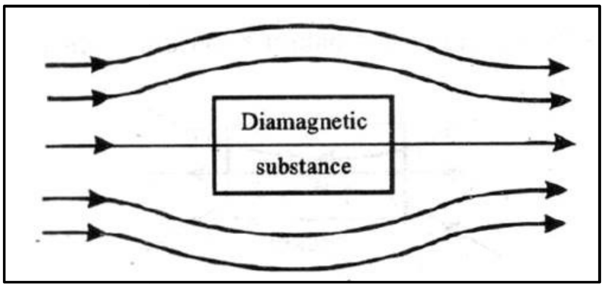
11. Moving coil Galvanometer.



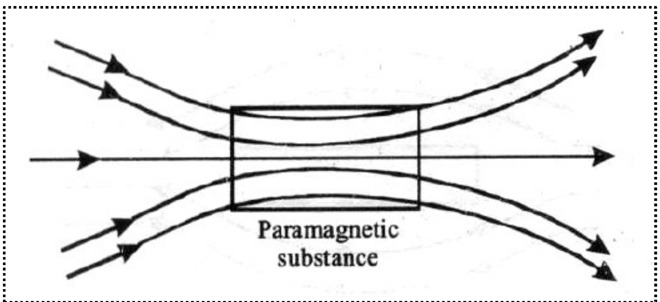
12. Magnetic field lines due to a finite solenoid has been shown below.



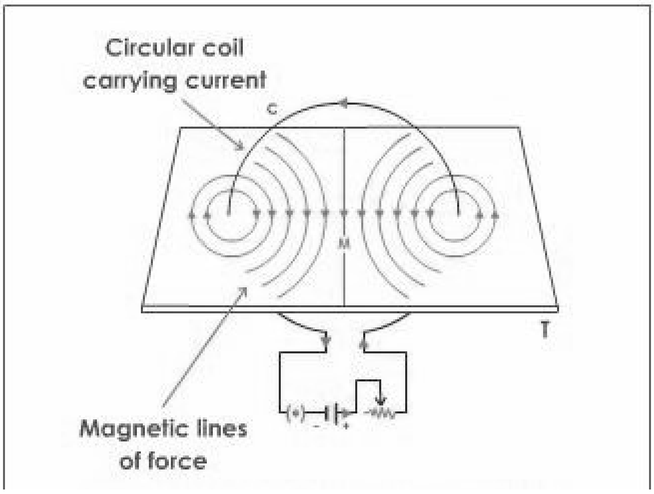
13. (i) Behavior of magnetic field lines when diamagnetic substance is placed in an external field.



(ii) Behavior of magnetic field lines when paramagnetic substance is placed in a external field.



14. Magnetic lines of force due to current carrying coil have been shown in the diagram given below.

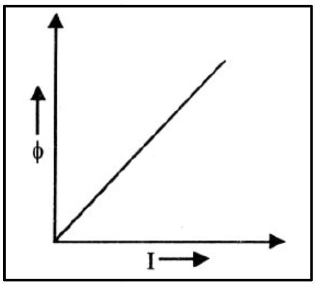


15. (i) Since $\phi = LI$

where, I = Strength of current through the coil at any time

ϕ = Amount of magnetic flux linked with all turns of the coil at the time

and, L= Constant of proportionality called coefficient of self induction

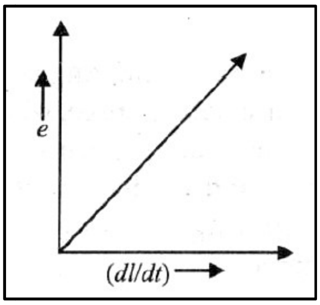


(ii) Induced emf,

$$e = -\frac{d\phi}{dt} = -\frac{d}{dt}(LI)$$

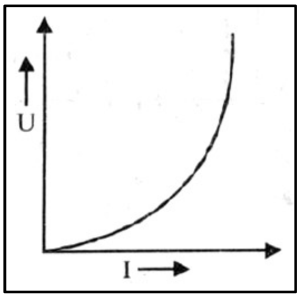
i.e., $e = -L\frac{dI}{dt}$

[drawn considering only magnitude of e]

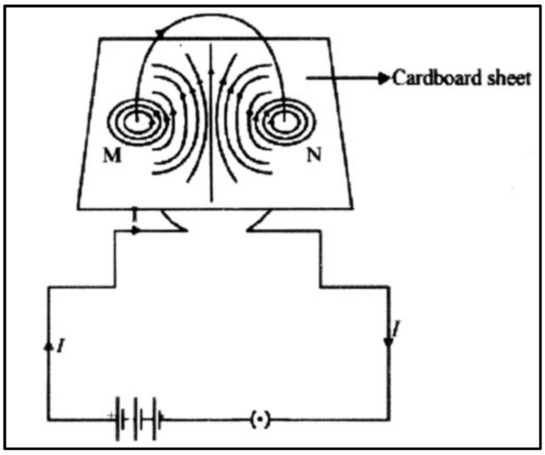


(iii) Since magnetic potential energy is given by,

$$U = \frac{1}{2} L I^2$$



16.



EMI, ALTERNATING CURRENT AND EM WAVES

17. Draw a schematic diagram of a step-up transformer.
[All India 2010, Delhi 2011C]

18. Sketch a schematic diagram depicting electric and magnetic fields for an electromagnetic wave propagating along the Z-direction.
[Delhi 2009]

19. An em wave is travelling in a medium with a velocity $\vec{v} = V\hat{i}$. Draw a sketch showing the propagation of the em wave, indicating the direction of the oscillating electric and magnetic fields. [Delhi 2013]

20. Draw a graph to show variation of capacitive-reactance with frequency in an a.c. circuit. [All India 2015]

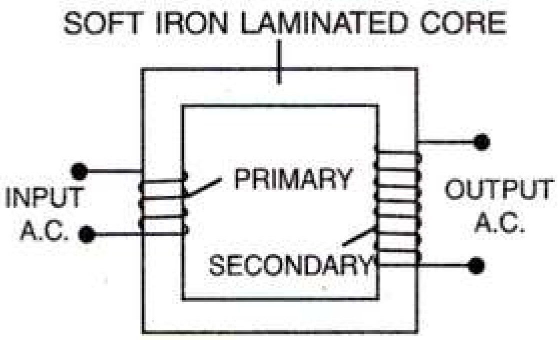
21. Draw a schematic sketch of the electromagnetic waves propagating along the + *x*-axis. Indicate the directions of the electric and magnetic fields.
[All India 2015]

22. Draw a labelled diagram of a step-down transformer.
[All India 2016]

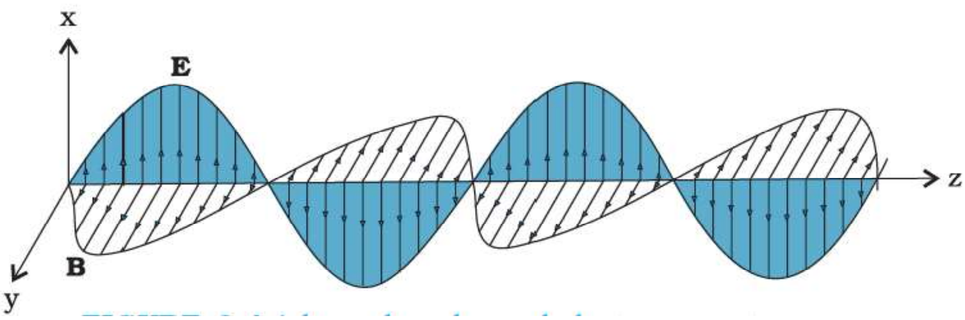
23. Draw a labelled diagram of an ac generator.
[All India 2017]

SOLUTIONS

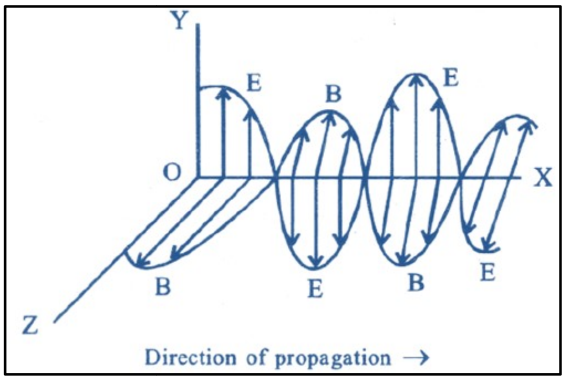
17. Schematic diagram of a step-up transformer.



18. A charge q oscillating at certain frequency produces an oscillating electric field in space, which produces an oscillating magnetic field. The oscillating electric and magnetic fields thus regenerate each other and produce electromagnetic wave. The frequency of the electromagnetic wave equals the frequency of oscillation of the charge.



19. From $\vec{V} = V\hat{i}$, it is clear that the wave is propagating along the x-axis. The direction of electric field is along the y-axis and that of the magnetic field along the z-axis.

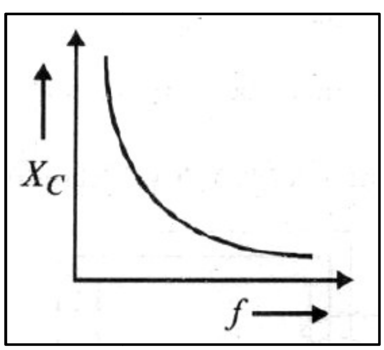


20. Capacitive reactance

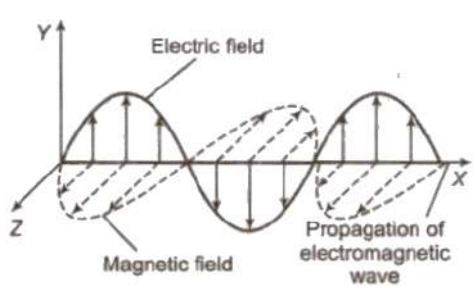
$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

Therefore, it is inversely proportional to frequency f.

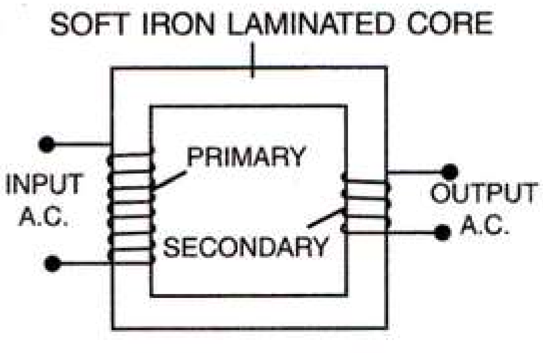
Graph: X_C versus f



21. Schematic sketch of the electromagnetic waves:

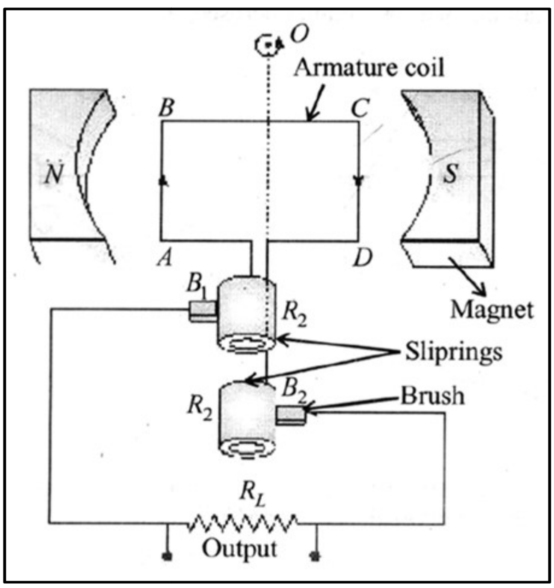


22. A labelled diagram of a Step-down transformer



23. (a) AC Generator: It is used to convert mechanical energy into electrical energy.

Principle: It works on the principle of electromagnetic induction.



OPTICS

24. Draw a neat labelled ray diagram of an astronomical telescope in normal adjustment.

[Delhi 2008, All India 2010, 2016, 2017]

25. Draw a schematic ray diagram of reflecting telescope showing how ray coming from a distant object are received at the eye-piece.

[Delhi 2008, 2016, Foreign 2010]

26. Draw a neat labelled ray diagram of a compound microscope.

[Foreign 2008, All India 2010, Delhi 2008, 2009, 2010, 2014]

27. Draw a ray diagram, showing the passage of a ray of light through a prism when the angle of incidence is 52°.

[Delhi 2010C]

28. A ray of monochromatic light is incident on one of the faces of an equilateral triangular prism of refracting angle A. Trace the path of ray passing through the prism.

[Foreign 2011]

29. An equiconvex lens of refractive index μ_1 focal length $'f'$ and radius of curvature 'R' is immersed in a liquid of refractive index μ_2. For (i) $\mu_2 > \mu_1$ and (ii) $\mu_2 < \mu_1$, draw the ray diagrams in the two cases when a beam of light coming parallel to the principal axis is incident on the lens.

[All India 2013C]

30. Draw a labelled ray diagram of a refracting telescope.

[All India 2013, Delhi 2013C]

31. Draw a diagram to show refraction of a plane wavefront incident on a convex lens and therefore draw the refracted wavefront.

[Delhi 2009]

32. Plot a graph showing the variation of stopping potential with the frequency of incident radiation for two different photosensitive materials having work function W_1 and $W_2 (W_1 > W_2)$.

[Delhi 2010]

33. Show the variation of photocurrent with collector plate potential for different intensities but same frequency of incident radiation.

[Foreign 2011]

34. Draw a graph between the frequency of incident radiation (n) and the maximum kinetic energy of the electrons emitted from the surface of a photosensitive material.

[Foreign 2012, Delhi 2014]

35. Show on a plot the nature of variation of photoelectric current with the intensity of radiation incident on a photosensitive surface.

[Delhi 2013C, 2014]

36. In Young's double slit experiment, plot a graph showing the variation of fringe width versus the distance of the screen from the plane of the slits keeping other parameters same.

[All India 2015]

37. Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials A and B having threshold frequencies$V_A > V_B$.
   (i) In which case is the stopping potential more and why?
   (ii) Does the slope of the graph depend on the nature of the material used? Explain.

[All India 2016]

38. Draw a graph showing the variation of intensity (I) of polarized light transmitted by an analyser with angle (θ) between polarizer and analyser.
[All India 2016]

39. Draw a proper diagram to show how the incident wavefront traverses through the lens and after refraction focusses on the focal point of the lens, giving the shape of the emergent wavefront.

[All India 2016]

40. Plot a graph showing variation of de-Broglie wavelength λ versus $\frac{1}{\sqrt{V}}$, where V is accelerating potential for two particles A and B carrying same charge but of masses $m_1, m_2 (m_1 > m_2)$.

[Delhi 2016]

41. Draw a graph showing variation of intensity in the interference pattern against position 'x' on the screen.

[Delhi 2016]

42. Plot a graph to show variation of the angle of deviation as a function of angle of incidence for light passing through a prism.

[Delhi 2016]

43. Draw the intensity pattern for single slit diffraction and double slit interference.

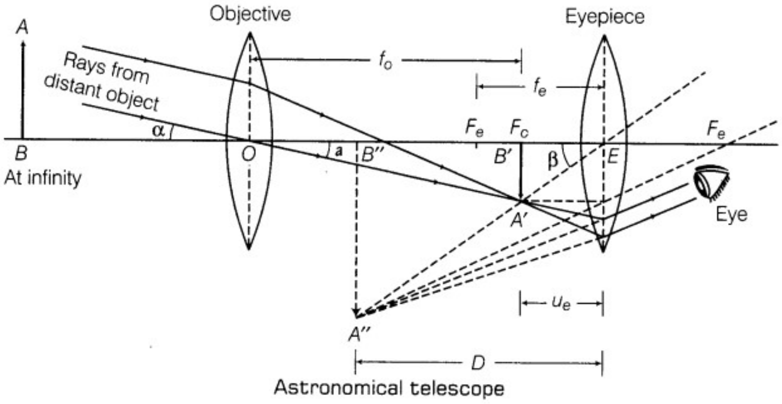
[All India 2017]

44. Draw a ray diagram to show the image formation by a combination of two thin convex lenses in contact.

[All India 2017]

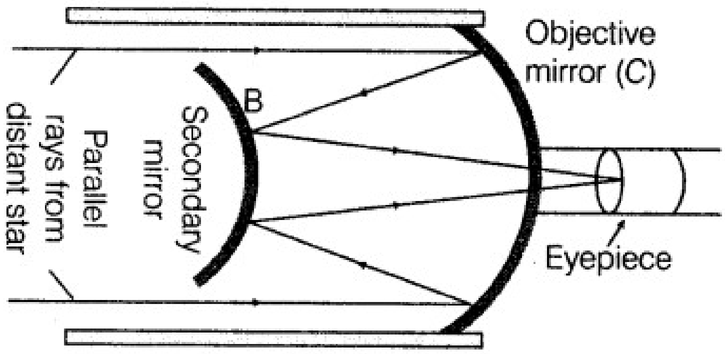
SOLUTIONS

24. Ray diagram of an astronomical telescope in normal adjustment.

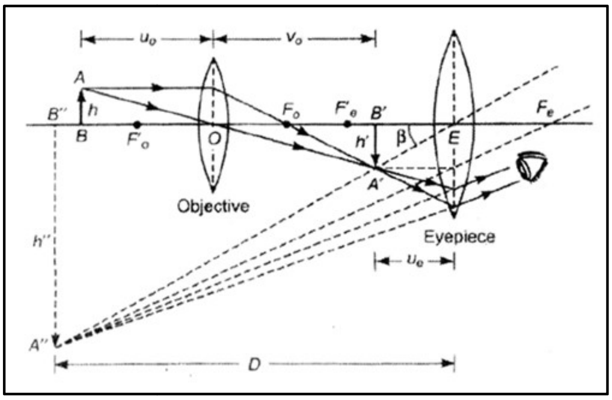


Astronomical telescope

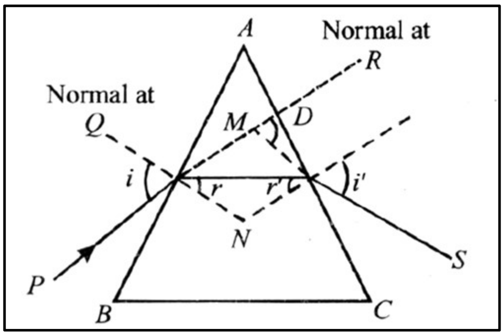
25. The ray diagram of reflecting telescope showing how ray coming from distant object are received at the eye-piece is shown in figure.



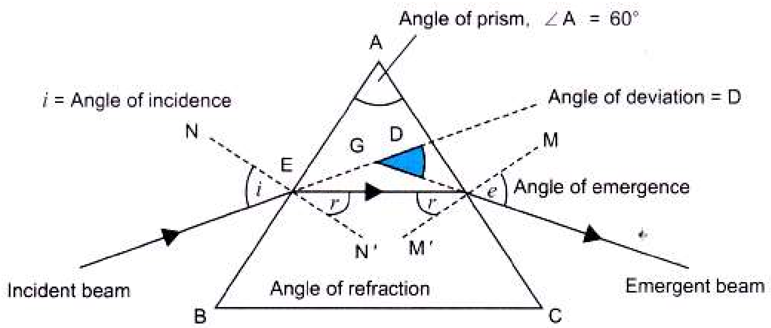
26. Ray diagram of a compound microscope:



27. The ray diagram in the condition of minimum deviation . Angle of incidence ∠i = 52°

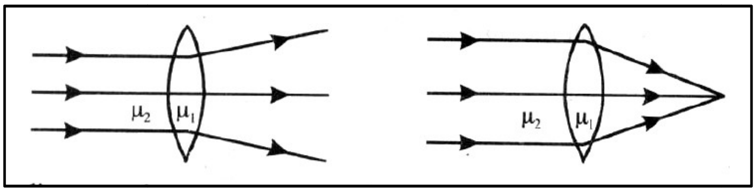


28. Ray passing through a prism.

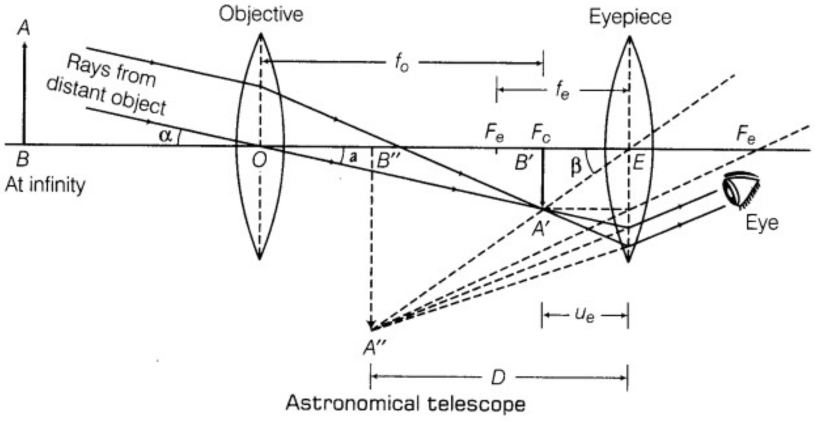


Refraction of light through a prism

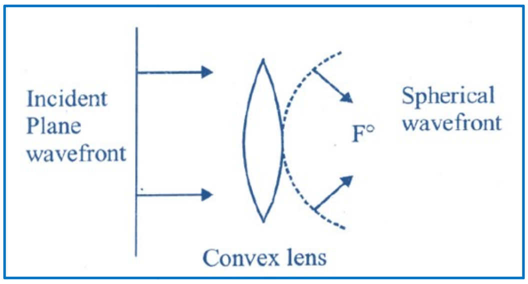
29. (i) Ray diagram: for $\mu_2 > \mu_1$ (ii) Ray diagram: for $\mu_2 > \mu_1$



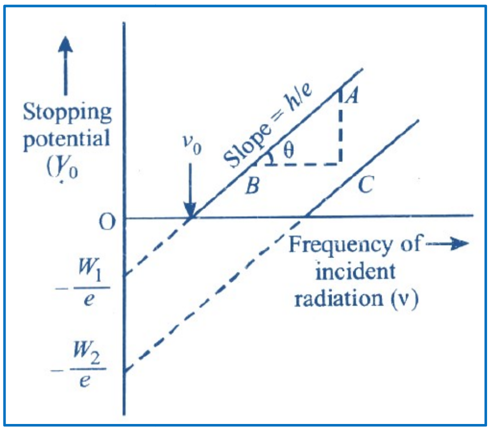
30. Ray diagram of a refracting type telescope.



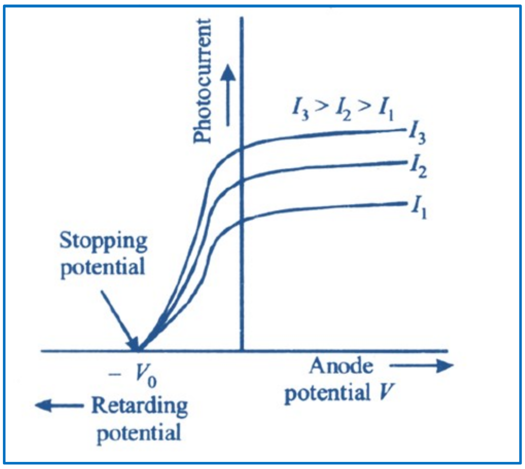
31. Diagram showing refraction of a plane wavefront



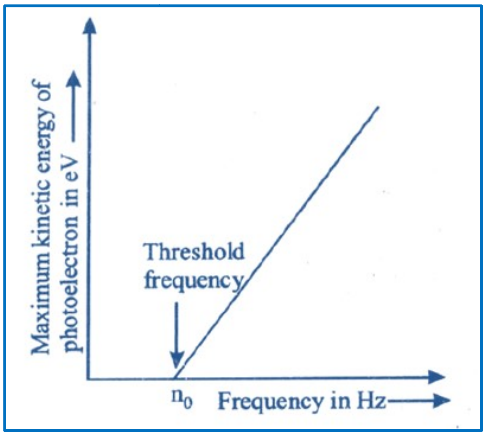
32. The variation of stopping potential with frequency of incident radiation .



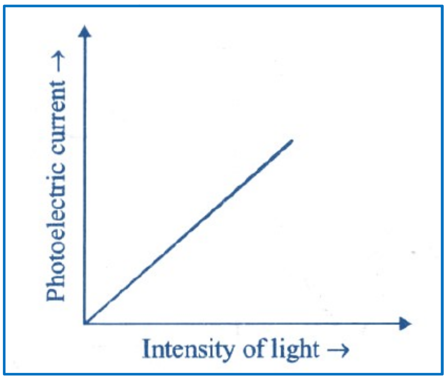
33. The variation of photocurrent for different intensities at constant frequency .



34. Graph between the frequency of incident radiation and the maximum kinetic energy of the electrons emitted.



35. Graph of photoelectric current versus intensity of light



36. The fringe width in Young's double slit experiment is given by

$\beta = \frac{\lambda D}{d}$

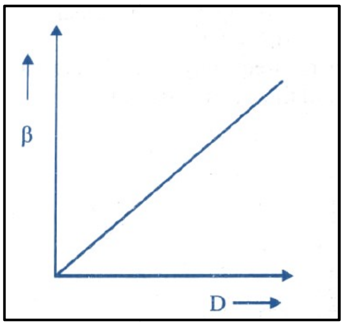
where λ = wavelength of source

D = distance between the slits and screen

d = distance between the slits

$\Rightarrow \beta \propto D$

The variation of fringe width with distance of screen from the slits



It is linear graph with slope equal to λ/d. So for the fringe width to vary linearly with distance of screen from the slits, the ratio of wavelength to distance between the slits should remain constant. Therefore, we take wavelengths (λ) of incident light almost equal to the width of the slit(β).

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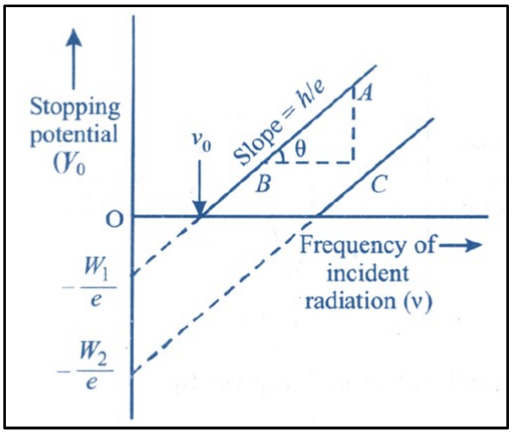
37. The variation of stopping potential with frequency of incident radiation for two photosensitive materials A and B having threshold frequencies



$v_A > v_B$.

From the graph, we see

(i) the stopping potential is inversely proportional to the threshold frequency, hence the stopping potential is higher for metal B.

(ii) the slope of the graph does not depend on the material used

As we know, from Einsteins photoelectric equation,

$K_{max} = hv - \phi_0 = eV_0$

Dividing by e, we get

$\frac{hv}{e} - \frac{\phi_0}{e} = V_0$

Hence the slope of the graph is $\frac{h}{e}$ (on comparing with the straight line equation) which is independent of the nature of the photoelectric material used.

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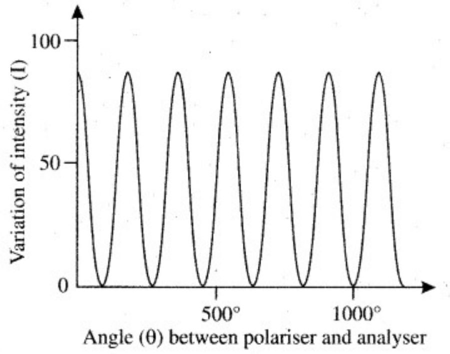
38. Variation of Intensity through Analyser



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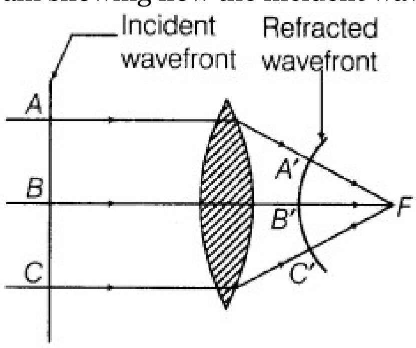
39. Diagram showing how the incident wavefront traverses through the lens.



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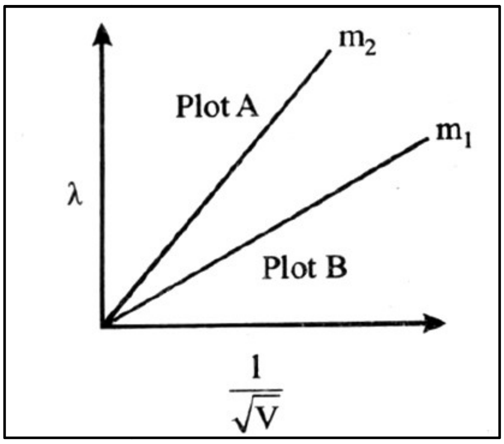
40. The de Brogile wavelength is given by $\lambda = \frac{h}{\sqrt{2mqV}}$ Where h- Planck's constant

The slope of the graph λ versus $\frac{1}{\sqrt{V}}$ is $\frac{h}{\sqrt{2mq}}$



The slope of the smaller mass (m_2) is larger; therefore, plot A in the above graph is for mass m_2

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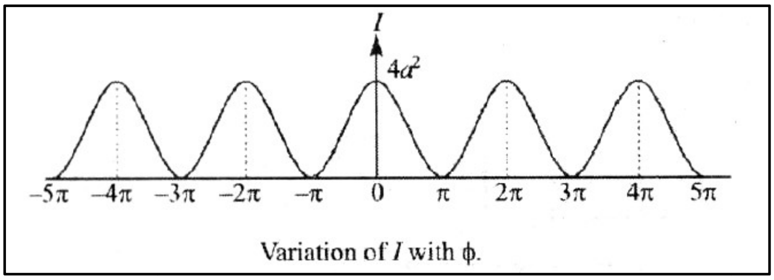
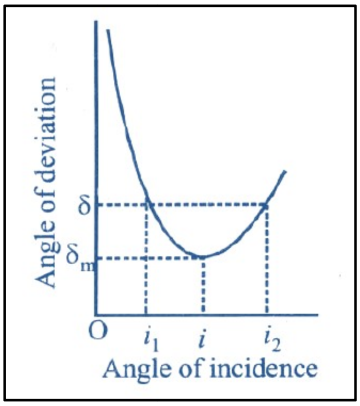
41. Graph of intensity distribution in Young's double-slit experiment

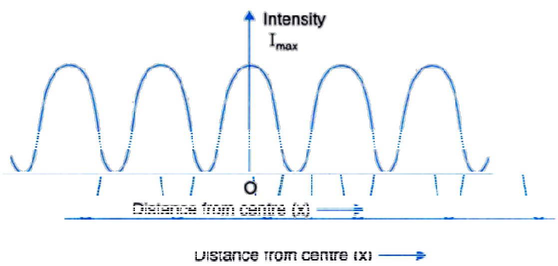


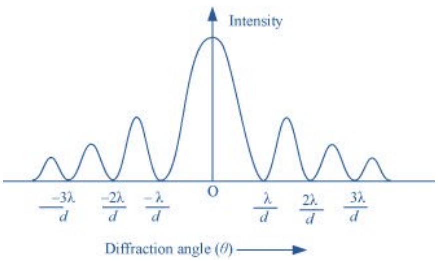
42. (i) If the angle of incidence is increased gradually, then the angle of deviation first decreases, attains a minimum value (δ_m) and then again starts increasing.



43. Intensity pattern for single slit diffraction

Intensity pattern for double slit diffraction





44. (a) Ray diagram to show the image formation by a combination of two thin convex lenses in contact:

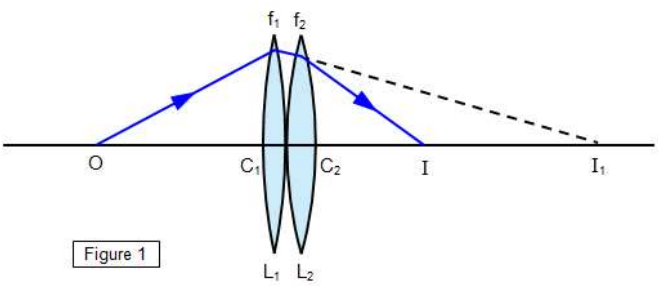


Figure 1

ATOMS AND NUCLEI

45. Draw a schematic arrangement of the Geiger-Marsden experiment for studying α-particle scattering by a thin foil of gold.
[All India 2009, Foreign 2010]

46. Draw the energy level diagram showing how the transitions between energy level result in the appearance of Lyman series.
[Delhi 2013]

47. Draw the energy level diagram showing how the line spectra corresponding to Paschen series occur due to transition between energy levels.
[Delhi 2013]

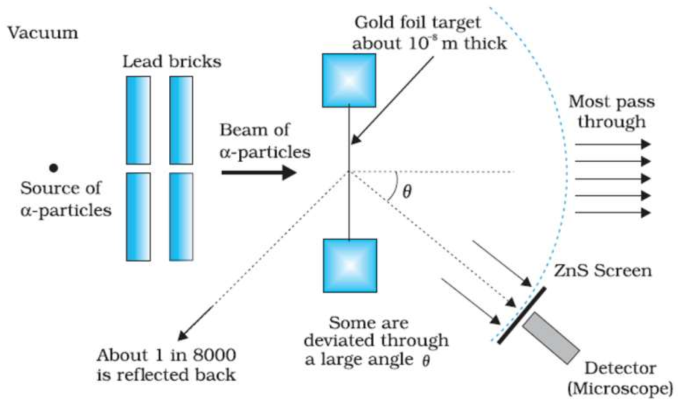
48. Draw the energy level diagram showing how the line spectra corresponding to Balmer series occur due to transition between energy levels.
[Delhi 2013]

49. Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei $20 > A > 240$.
[Foreign 2008, All India 2009C, 2010, 2013]

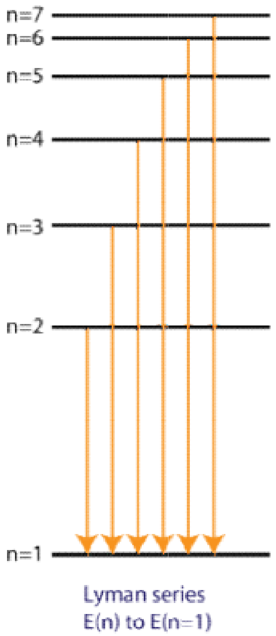
50. Draw a plot of potential energy between a pair of nucleons as a function of their separation. Mark the regions where potential energy is (i) positive and (ii) negative.
[Delhi 2013]

SOLUTIONS

45. Given figure shows a schematic diagram of Geiger-Marsden experiment.

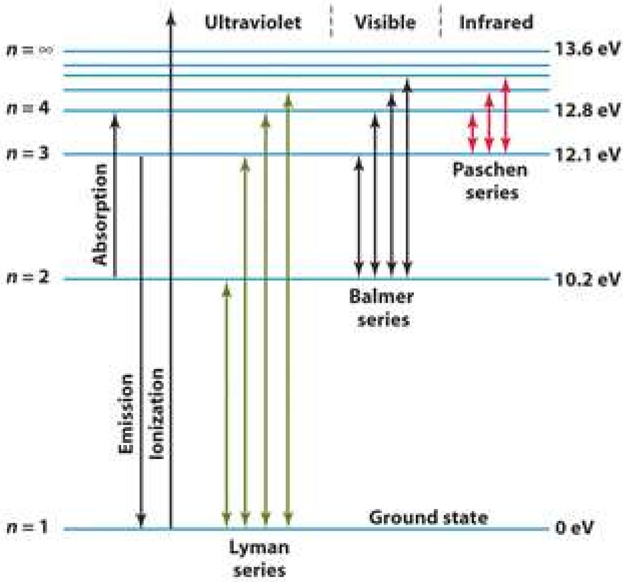


46. Energy level diagram for the Lymann series:



The Lymann series of the hydrogen atom is produced when transitions take place from higher orbit to the first orbit $n_1 = 1$

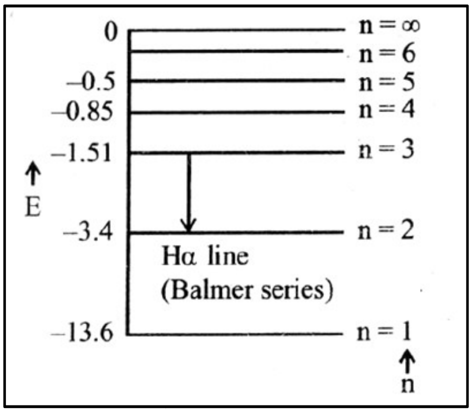
47. Energy level diagram for the Paschen series



The Paschen series of the hydrogen atom is produced when transitions take place from higher orbits to the third orbit.
i.e., $n_1 = 3$ and $n_2 = 4, 5, 6, \ldots \ldots \ldots$so on.

---

48. Energy level diagram: The Balmer series is produced when transition take place from higher orbits to n = 2 as shown in the figure.



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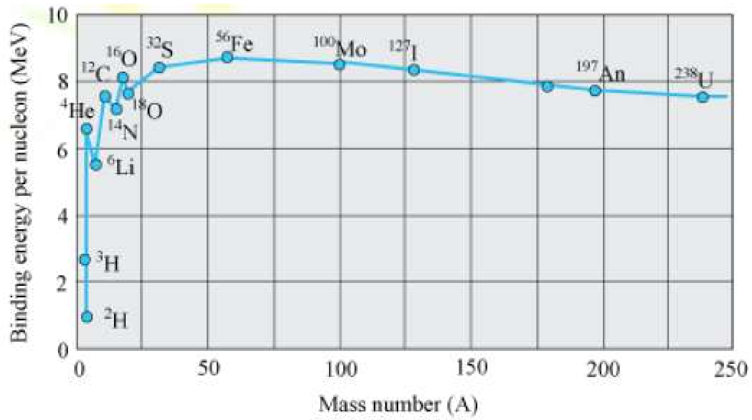
49. The binding energy per nucleon curve .



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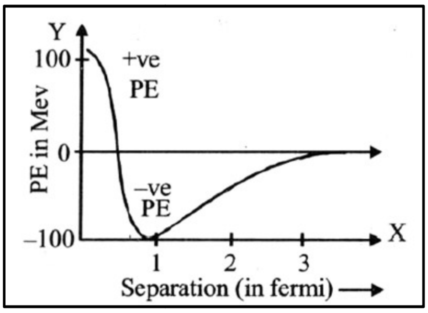
50. Plot of potential energy between a pair of nucleons as a function of their separation:

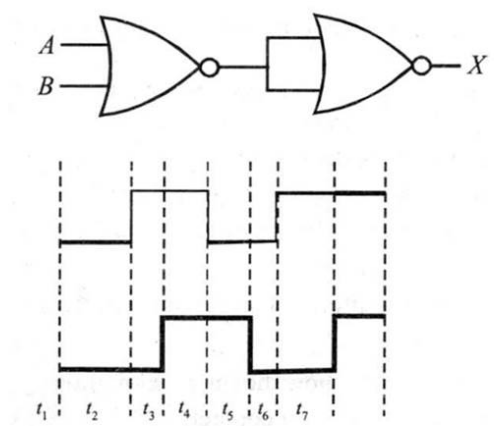


ELECTRONICS AND COMMUNICATION SYSTEMS

51. Draw the circuit diagram for studying the characteristics of an *n-p-n* transistor in common emitter configuration. Sketch the typical (i) input and (ii) output characteristics in CE configuration

[All India 2008, 2014, Delhi 2009, 2009C, 2010C, 2013C, Foreign 2012]

52. Draw the output waveform at X, using the given inputs, A and B for the logic circuit shown below.



[Delhi 2008, 2011]

53. Draw a simple circuit of a CE transistor amplifier.

[Foreign 2008, Delhi 2012]

54. Draw V-I characteristics of a *p-n* junction diode.

[All India 2009, 2011, 2013]

55. Draw the logic circuit of a NAND gate.

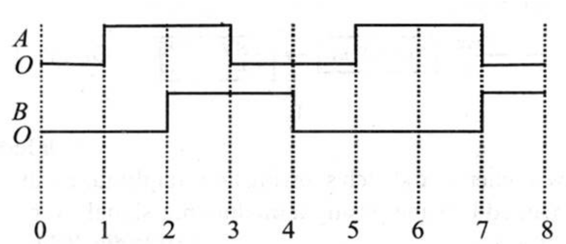
[All India 2009, Foreign 2011]

56. Draw the logic circuit of AND gate.

[All India 2009, Foreign 2011]

57. Draw I-V characteristics of zener diode. [Delhi 2009]

58. Sketch the output wavefrom from an AND gate for the input *A* and *B* shown in the figure.



[Delhi 2009]

59. Draw the circuit diagram showing how a *p-n* junction diode is (i) forward biased, (ii) reverse biased

[All India 2010, 2011c]

60. Draw the circuit diagram of an illuminated photodiode in reverse bias.
[Delhi 2010]

61. Draw the logic circuit of a NOT gate.

[Foreign 2011]

62. Show how intensity of current varies with illumination intensity in a photodiode.

63. Plot a graph showing variation of current versus voltage for the material GaAs.

[Delhi 2014]

64. Draw a labelled block diagram of a simple modulator for obtaining an AM signal.

[All India 2008, Delhi 2008C, 2009, Foreign 2010]

65. Draw a plot of the variation of amplitude versus ω for an amplitude modulated wave.

[Delhi 2008]

66. Draw block diagram of a detector for AM waves?

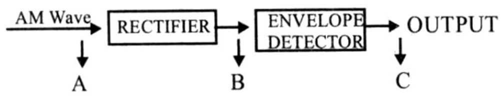
[Delhi 2008C, 2013C]

67. Draw a schematic diagram showing the
(i) ground wave (ii) sky wave and (iii) space wave propagation modes for em waves.

[All India 2011C, Delhi 2012]

68. Draw diagram for following:
(a) Carrier wave
(b) Modulating wave
(c) AM wave
(d) FM wave
(e) PM wave

69. Figure shows the block diagram of a detector for AM signal. Draw the waveforms for the (i) input AM wave at A (ii) output B at rectifier and (iii) output signal at C.



[All India 2013C]

70. Draw a schematic sketch showing how amplitude modulated signal is obtained by superposing a modulating signal over a sinusoidal carrier wave.
[All India 2014, Delhi 2013]

71. Draw the necessary energy band diagrams to distinguish between conductors, semiconductors and insulators. How does the change in temperature affect the behavior of these materials? Explain briefly.
[All India 2015]

72. Draw a circuit diagram of a transistor amplifier in CE configuration.
[Delhi 2015]

73. Draw the circuit diagram of a half wave rectifier.

[All India 2013C, 2016]

74. Draw the circuit diagram for studying the input and output characteristics of *n-p-n* transistor in common emitter configuration. [Delhi 2016]

75. Draw the circuit diagram of a full wave rectifier.

[All India 2017]

76. Draw a block diagram of a generalized communication system.

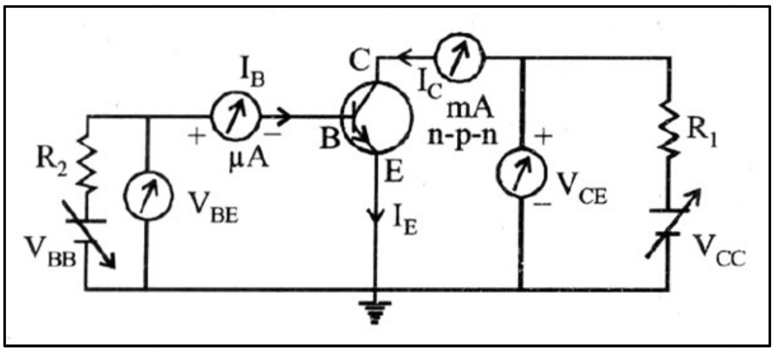
[All India 2017]

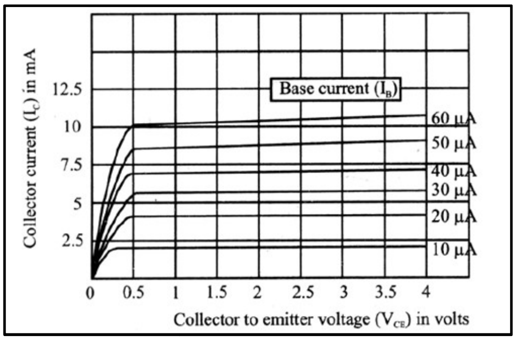
SOLUTIONS

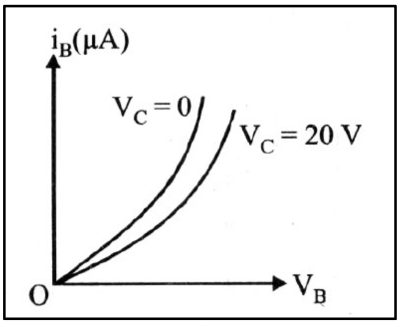
51. The circuit diagram of n-p-n transistor in CE configuration is as shown below

(i) Input characteristics: It is a graph between base voltage V_B and the base current I_B for a constant value of collector voltage V_C.

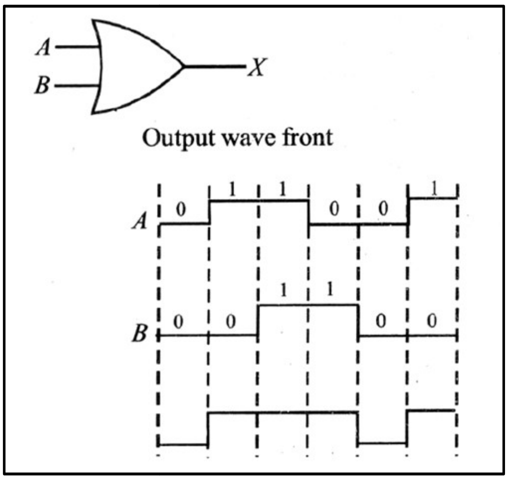
(ii) Output characteristics: It is a graph between collector voltage V_{CE} and collector current Ic for a constant value of base current I_B.



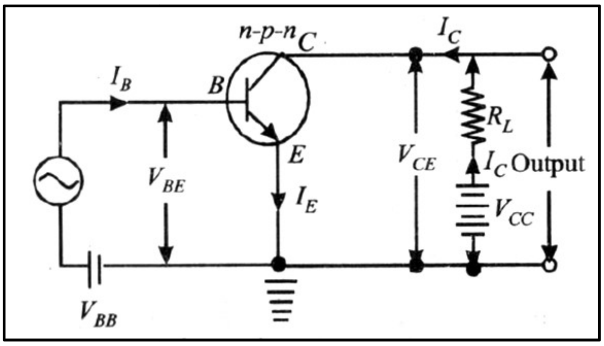




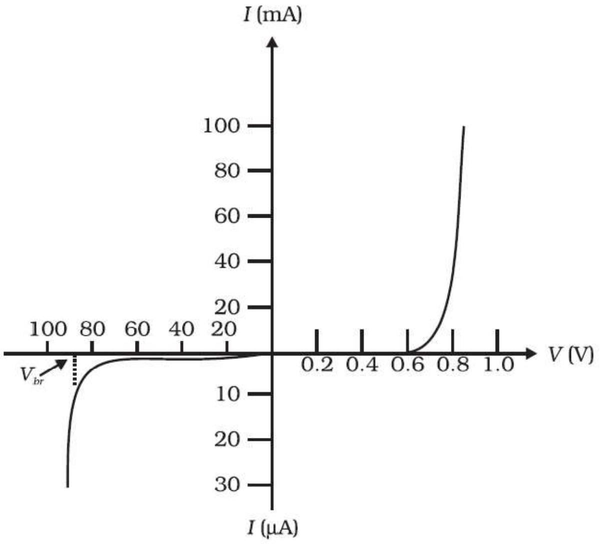
52. Equivalent gate is OR gate.
Logic symbol of OR gate and the output waveform .
Symbol of OR gate.



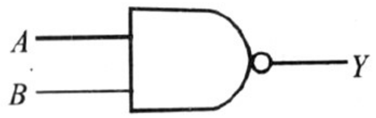
53. Circuit diagram of a common emitter transistor amplifier



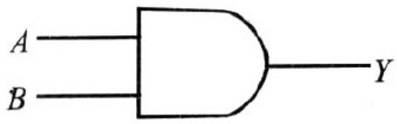
54. *V-I* characteristic of a *p-n* junction diode.



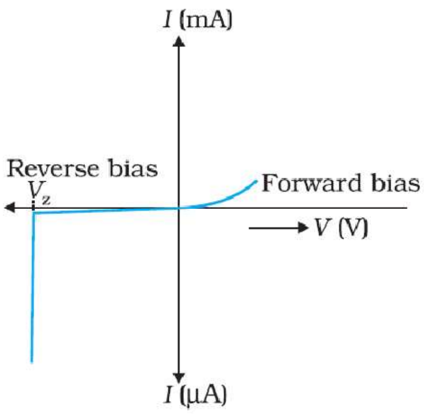
55. Logic circuit of NAND gate.



56. Logic circuit of a AND gate

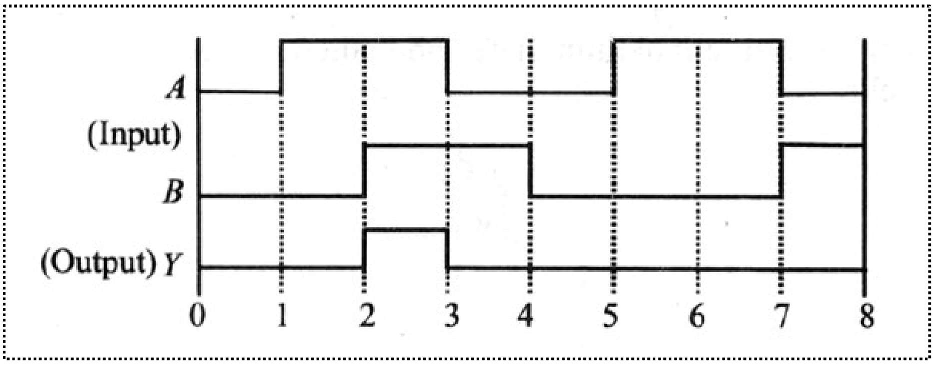


57. Zener diode is operated in the reverse breakdown region. The voltage across it remains constant.
V-I characteristic of a zener diode



58. The output for an AND gate is given as

For 0 to 1, A = 0, B =0, therefore, y = 0
For 1 to 2, A = 1, B =0, therefore, y = 0
For 2 to 3, A = 1, B =1, therefore, y = 1
For 3 to 4, A = 0, B =1, therefore, y = 0
For 4 to 5, A = 0, B =0, therefore, y = 0
For 5 to 6, A = 1, B =0, therefore, y = 0
For 6 to 7, A = 1, B =0, therefore, y = 0
For 7 to 8, A = 0, B =1, therefore, y = 0



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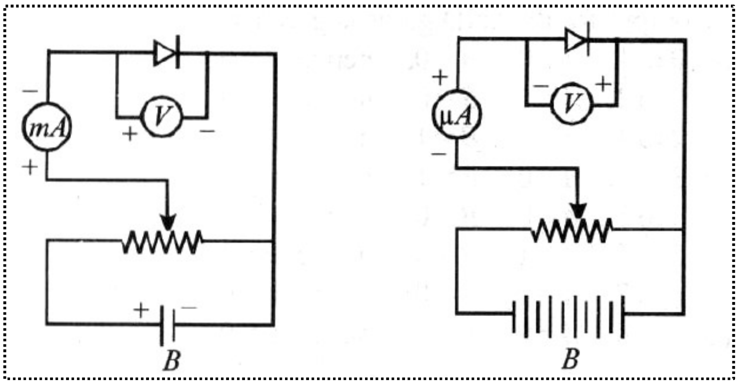
59. Circuit diagram of (i) forward biased and (ii) reverse biased p-n junction diode .

(i) Circuit diagram of forward biased p-n junction diode.

(i) Circuit diagram of reverse biased p-n junction diode



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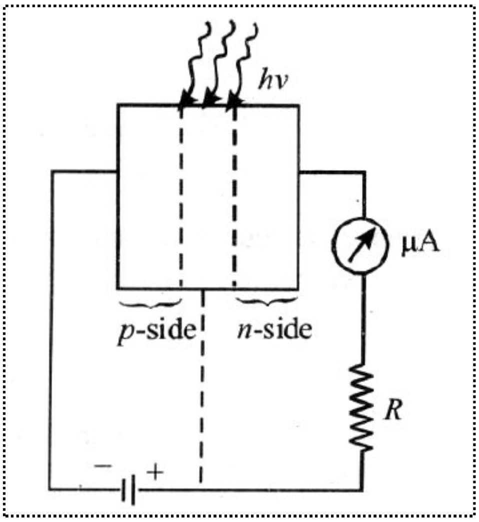
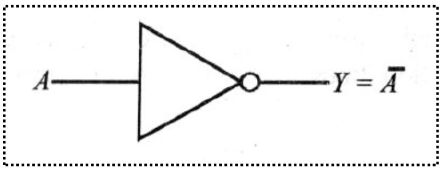
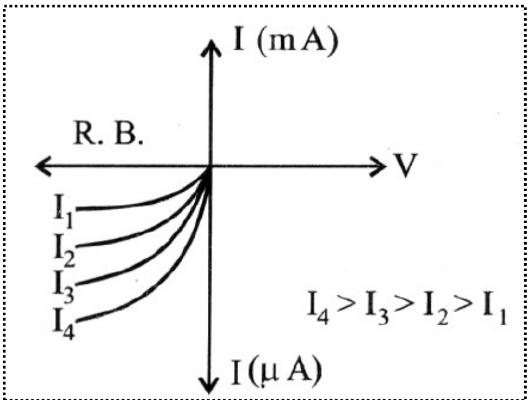
60. Circuit diagram of illuminated photodiode in reverse bias .



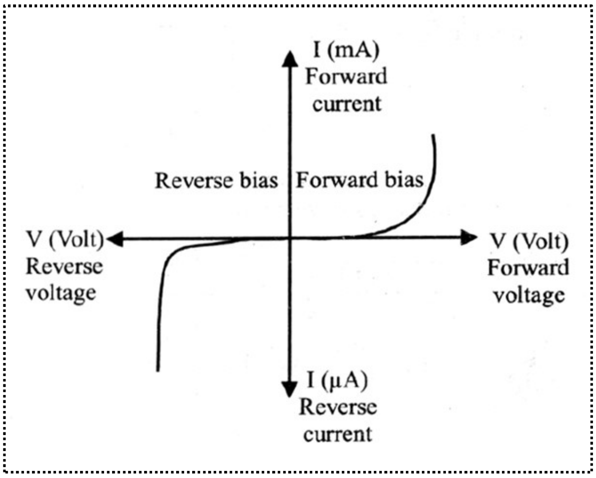
61. Logic circuit of a NOT gate.



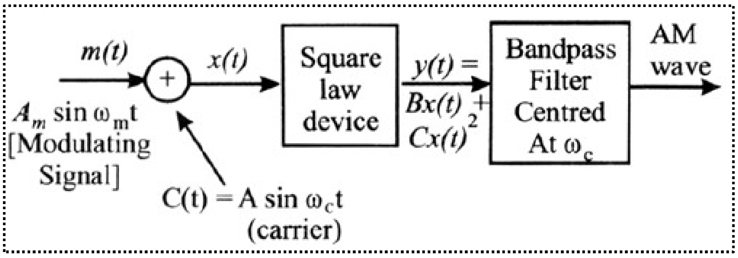
62. Graph showing variation of intensity of current with illumination intensity.



63. Current – Voltage characteristic graph for GaAs:



64. Block diagram of a simple modulator



65. A plot of amplitude versus ω for an amplitude modulated wave.

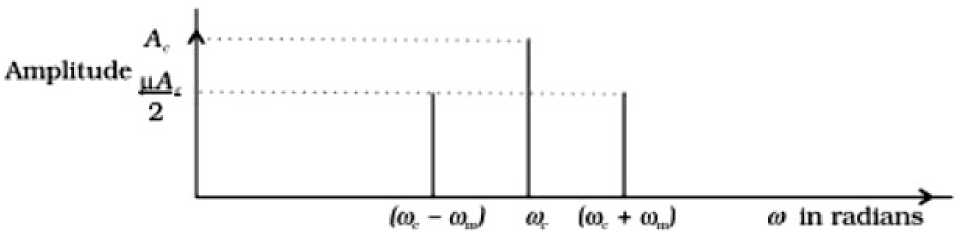
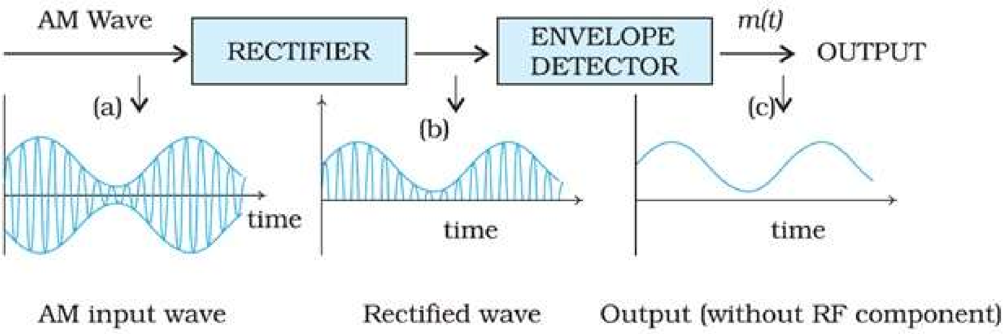


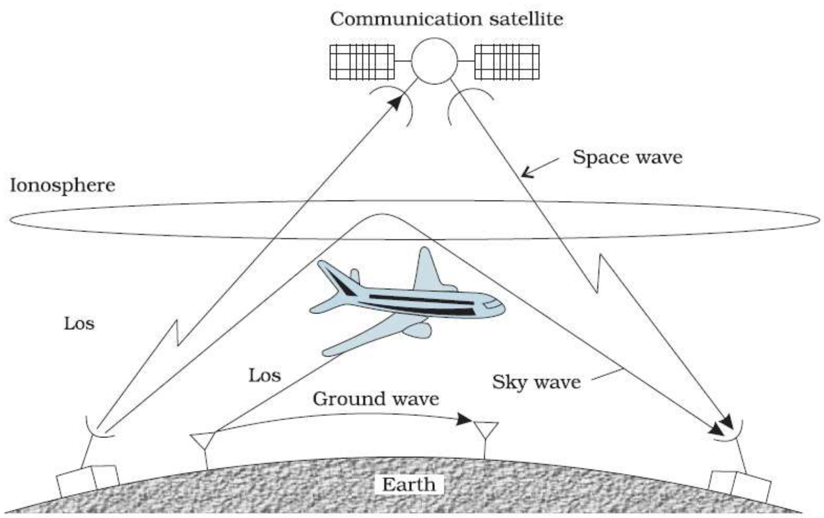
FIGURE 15.9 A plot of amplitude versus ω for an amplitude modulated signal.

66. The block diagram is as shown below:



67. Schematic diagram of showing propagation of

(i) Ground wave (ii) Sky wave (iii) Space wave



68.

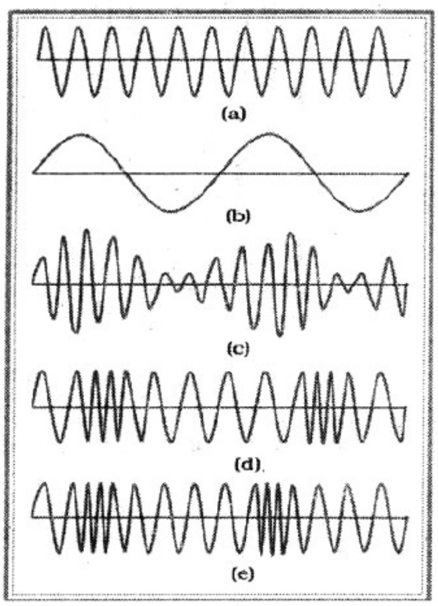
(a) Carrier wave

(b) Modulating wave

(c) AM wave

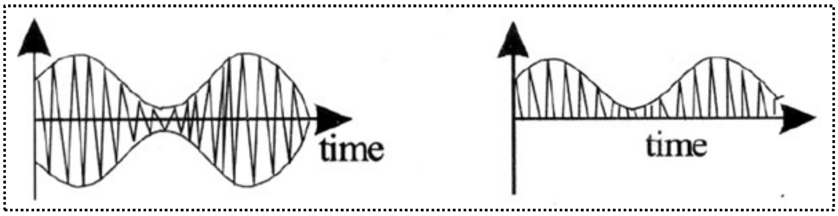
(d) FM wave

(e) PM wave

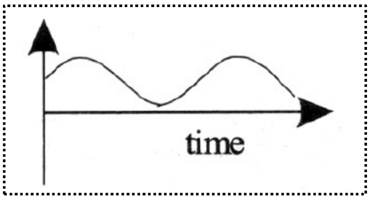


69. Wave forms for the:

(i) Input AM wave at A (ii) Output B at the rectifier



(iii) Output signal at C



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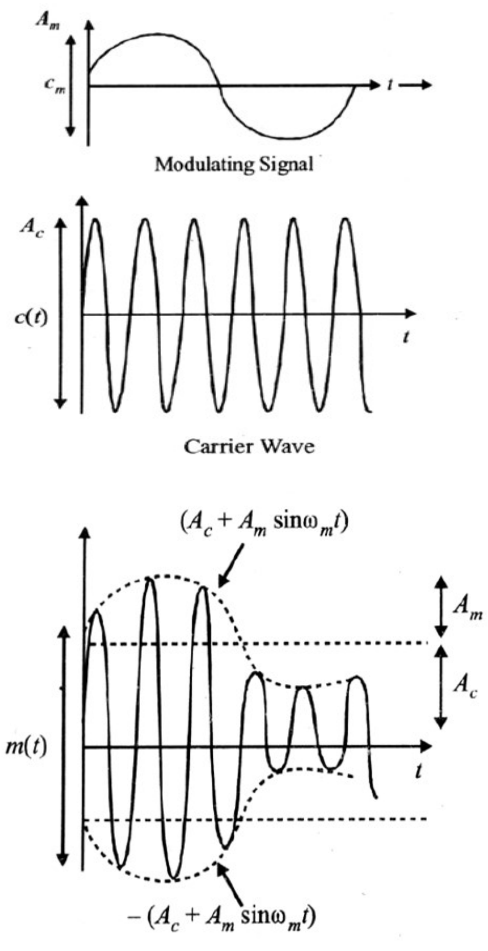
70. Amplitude modulated signal is obtained by superposing a modulating signal over a sinusoidal carrier wave is shown in the figure given below



Amplitude Modulated Wave

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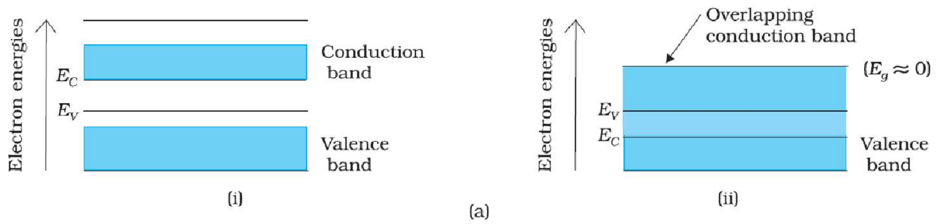
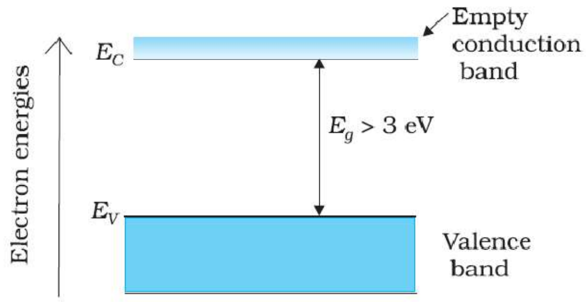
71. In conductors, the conducting and the valence band overlap each other. As the temperature increases, the conductivity of the conductors decreases due to increase in the thermal motion of the free electrons.

Energy band diagram for a conductor

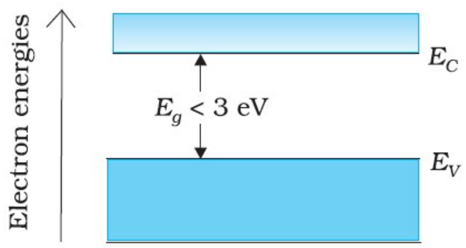


<u>Energy band diagram for an insulator</u>



The valance band is completely filled and the conduction band is empty. The energy band gap of the insulator is quite large. So, on increasing the temperature, the electrons of the valence band are not able to reach the conduction band. Therefore, electrical conduction in these materials is almost impossible.

<u>Energy band diagram for a semiconductor</u>



In semiconductor, the valence band is fully filled and the conducting band is empty but the energy gap between conduction band and valence band is quite small. At zero K, electrons are not able to cross even this small energy gap and, therefore, the conduction band remains totally empty.

At room temp., some electrons in the valence band acquire thermal energy greater than energy band gap, which is < 3 eV and jump over to the conduction band where they are free to move under the influence of even a small change in the temperature. As the temperature increases more and more electrons crosses the band gap and therefore conductivity increases.

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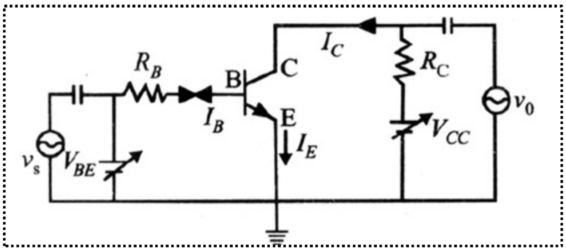
72. The circuit diagram of an n-p-n transistor amplifier in CE configuration is given below:



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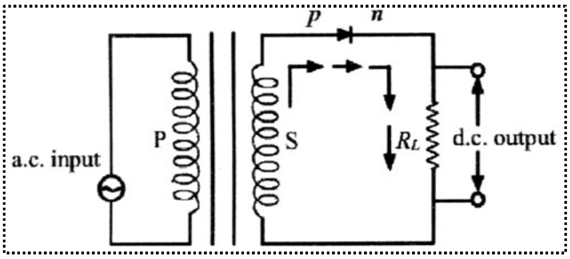
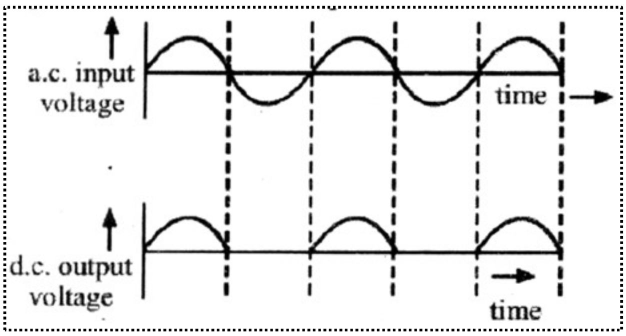
73. The circuit diagram for a half wave rectifier is shown below:



Working: During the positive half cycle of the input a.c., the p-n junction is forward biased i.e., the forward current flows from p to n and the diode offers a low resistance path to the current. Thus, we get output across-load i.e. a.c input will be obtained as d.c output.



During the negative half cycle of the input a.c., the p-n junction is revered biased i.e., the reverse current flows from n to p, the diode offers a high resistance path to the current. Thus, we get no output across-load.

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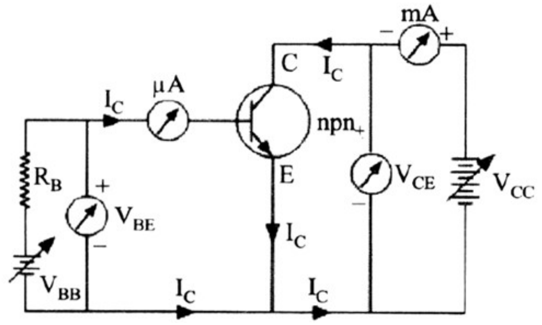
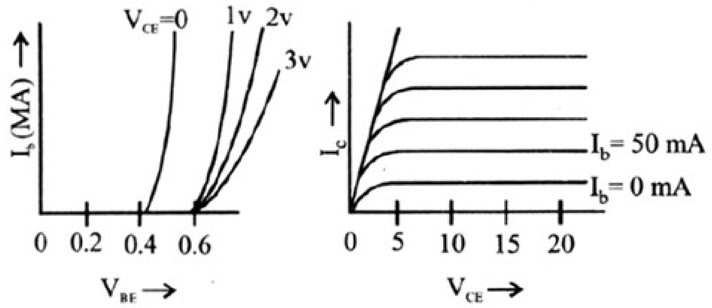
74. Circuit diagram for studying the input and output characteristic
Input characteristic. Curve is drawn between base current I_b and emitter



base voltage V_{EB}at constant collector emitter voltage V_{CE}
Output characteristic. Variation of collector current I_C with V_{CE} can be noticed for V_{CE} between 0 to 1 volt only.



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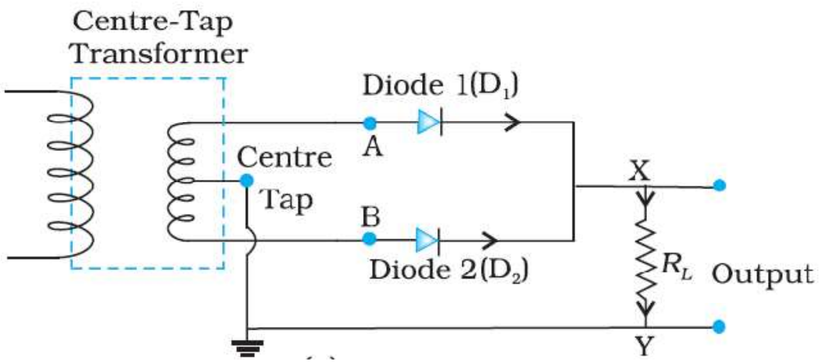
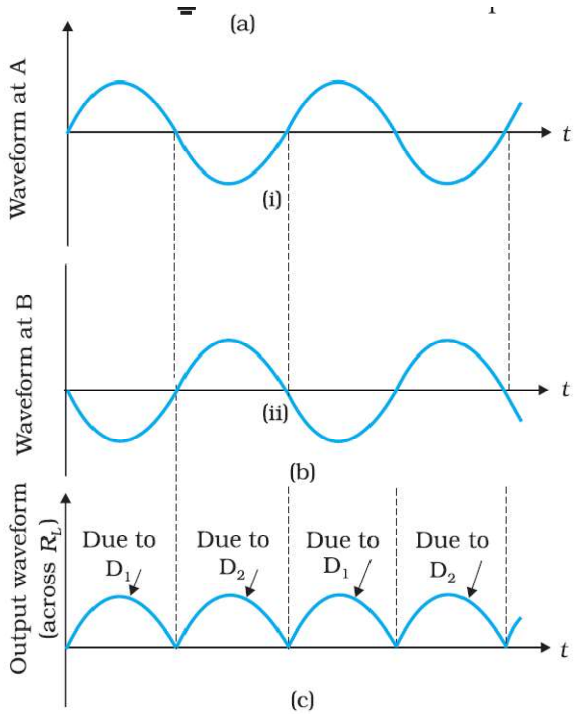
75. P-N junction diode as a full wave rectifier: The circuit uses two diodes connected to the ends of a centre tapped transformer. The voltage rectified by the two diodes is half of the secondary voltage i.e., each diode conducts for half cycle of input but alternately so that net output across load comes as half sinusoids with positive values only.



For positive cycle diode D_1 conducts (FB) but D_2 is being out of phase is reverse biased and does not conduct. Thus output across R_L is due to D_1 is R.B. but D_2 is F.B. and conducts as with respect to centretap point A is negative but B is positive. Therefore output across R_L is due to D_2.



76. Block diagram of a generalized communication system: